EQN

STACKCALC SC-32 SCIENTIFIC RPN CALCULATOR

4-Register RPN Stack Engine, Dual Stand Modes & Hardware Guide

SERIES 1 • CARD 09 / 12 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES

CARD 09 / 12 • SC-32-HW

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

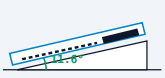
4-REGISTER RPN ENGINE & DUAL STANDS

Reverse Polish Notation (RPN) with 4 registers.
Numbers push onto stack; operators evaluate at once:
Stack Architecture: Regs T, Z, Y, X
Zero parentheses, zero ambiguity, zero errors.

CALIBRATED 32-KEYCAP TACTILE HARDWARE

- Display: 128×64 OLED with live 4-stack view
- Keycaps: 43 sculpted Shore 90A TPU keys over tactile switches
- Stack: Push on [ENTER], pop on op, LastX saved

OPERATING MODES: 11.6° DESK INCLINE & 78° PUZ



DESK STAND (11.6°)
Seated typing • Glare-free OLED



PUZZLE STAND (78°)
3-pc easel • Knocks flat to 6mm

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC	COLOR
SC-32 Calculator	1x	128×64 OLED	Obsidian
TPU Bumper Sleeve	1x	Shore 95A TPU	Black TPU
3-Piece Stand	1x	78° Easel	Matte Charcoal
Desk Stand Caddy	1x	11.6° Incline	Matte Charcoal
USB-C Cable	1x	1.0m Braided	Stealth Black
Mailer Box	1x	6×4 In Case	Kraft Box

TOTAL: Calculator + TPU Sleeve + Stands + Cable + Box = 6 Parts

POWER CONTROL & ANNUNCIATORS

- Instant Power Off: Tap [f] then [ON/OFF] to shut down.
- Auto-Sleep Standby: Sleeps after 5 min idle (<15 μA).
- Non-Volatile Memory: Registers (X, Y, Z, T) persist.
- Annunciators: RAD/DEG, [f], [g], battery on OLED header.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Press [f] [OFF] to turn off, or let auto-sleep preserve your stack. Dock in the Desk Stand or slip into the TPU bumper sleeve and drop in your backpack with zero loose parts!

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Any algebraic expression can be evaluated unambiguously without parentheses using a postfix stack.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: 4-REGISTER RPN STACK & EXPRESSION TREES

Step-by-Step Hands-On Tutorial: Pythagorean Hypotenuse $\sqrt{(3^2 + 4^2)} = 5$

SERIES 1 • CARD 09 OF 12 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES

CARD 09 / 12 • SC-32-HW

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/sc32-calculator/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS MP5 (Use Tools St...)
- CCSS MP6 (Attend to Precision)
- TEKS 111.39.c.1.C (Manipulative...)

Grades 8–16 STEM, Physics & Engineering

GUIDING INQUIRY

Where does the first number wait safely on the stack while you are entering the second number?

1 STEP 1: ENTER 3 AND 4 ONTO STACK

Power on the SC-32. Key: 3 [ENTER] 4 [ENTER].
Stack state: Register Y holds 3.0000; Register X holds 4.0000.
In RPN, numbers always enter before operators!

2 STEP 2: SQUARE BOTH REGISTERS

Press [⁻¹] [x²] on 4: Register X becomes 16.0000.
Press swap [x⇄y]: Register X becomes 3.0000; Y holds 16.0000.
Press [⁻¹] [x²] on 3: Register X becomes 9.0000.

3–4 STEPS 3 & 4: COLLAPSE STACK & SQRT

3. ADD SQUARES: Press [+]. The stack drops: 16.0000 + 9.0000 = 25.0000 in X!
4. SQUARE ROOT: Press [√x]. Result: 5.0000!
Combined Keystrokes: 3 [⁻¹] [x²] 4 [⁻¹] [x²] [+] [√x] → 5.

5 CORE PRINCIPLE: THE FREEDOM FROM PARENTHESES

Evaluating $\sqrt{(3^2 + 4^2)}$ in traditional algebraic mode requires nested parentheses. In RPN, you calculate from the inside out, exactly as you think! Intermediate numbers sit patiently on the stack until an operator consumes them. Zero parentheses, zero syntax errors!

★ EXTENSION INVESTIGATION: KINETIC ENERGY: $E = \frac{1}{2} m v^2$

Calculate kinetic energy for mass m=12 and velocity v=5: 0.5 [ENTER] 12 [×] 5 [⁻¹] [x²] [×]
→ 150.0000! Zero parentheses! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.